

Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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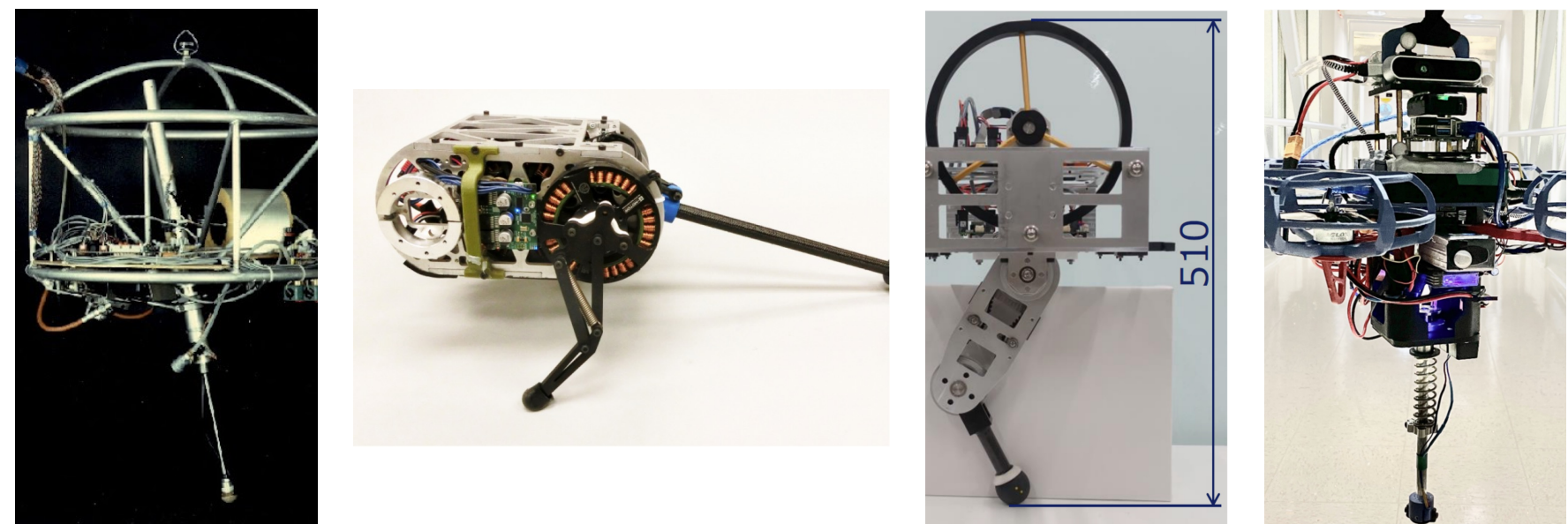
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1. Problem - hybrid dynamics create an authority gap

**Conventional Hopper** → **Inertial Appendages** → **Propeller Assistance**

Stance

Ground reaction force exists only during a short contact interval.

Flight

No contact: attitude becomes underactuated.

Disturbance

Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2. First proof - stable on 30° grass



Propellers off

After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

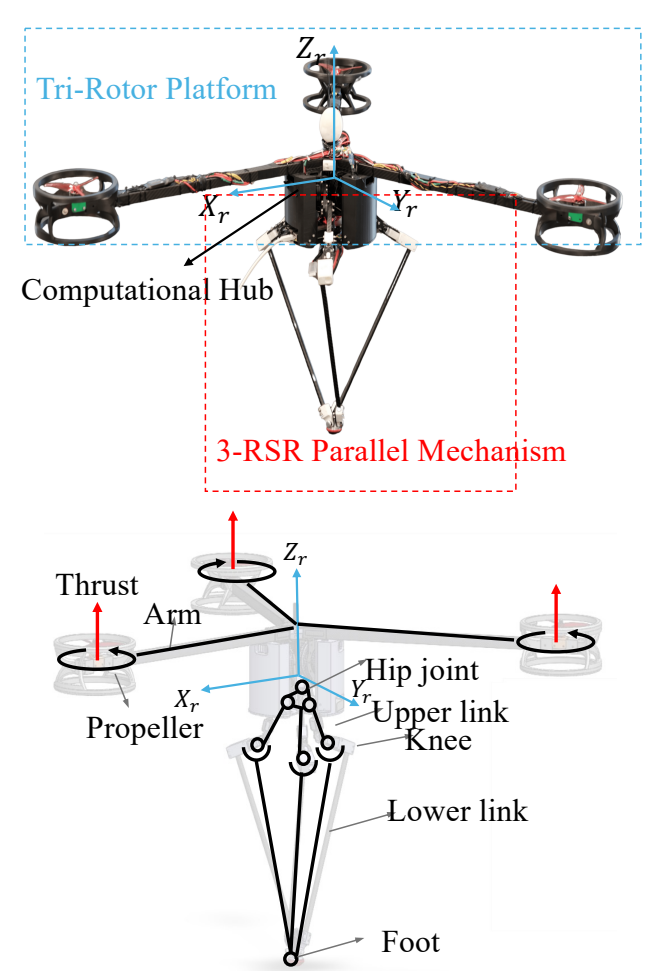
Propellers on

Differential thrust remains available through flight and restores attitude before the next landing.

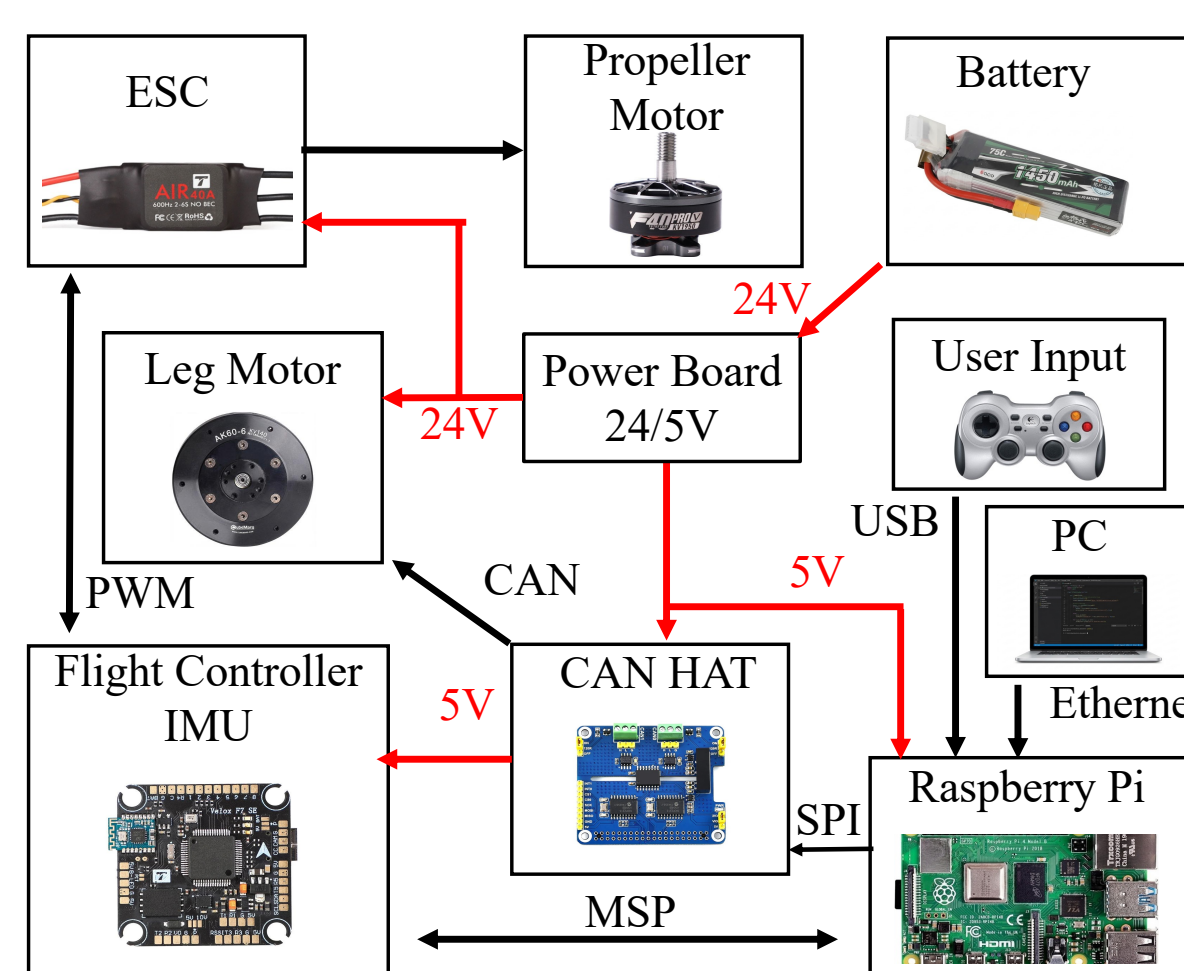
What this establishes

Residual authority in stance.**Primary** attitude authority in flight.**Faster recovery** from uneven impacts.**Not primary lift:** the rotors close the attitude-authority gap.**Propellers off:** attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

3. Platform - hardware built for leg-first hopping



Paper Fig. 2 - robot design.



Paper Fig. 3 - electronic architecture.

3-RSR parallel leg

Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.

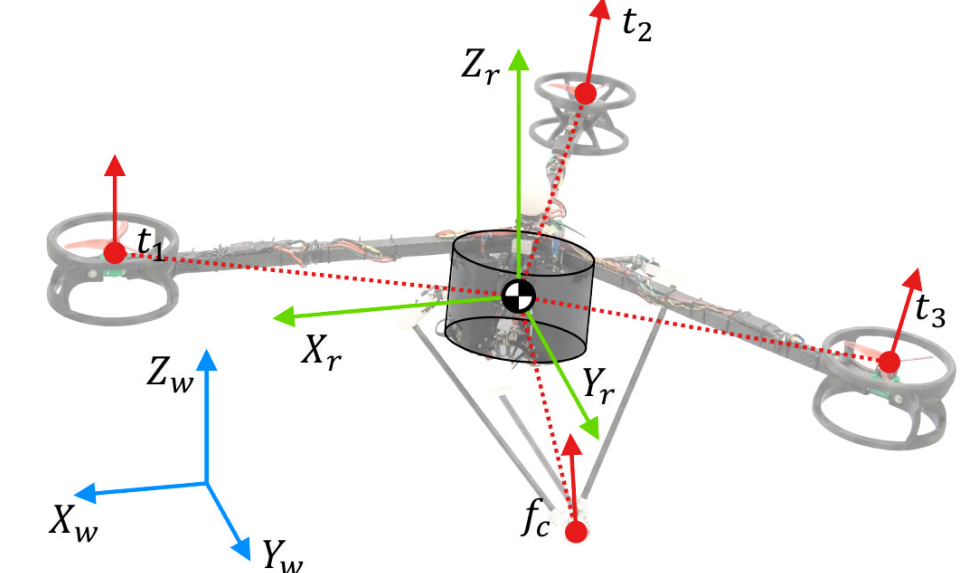
Computational hub

Batteries and electronics near the hip: low center of mass, low inertia.

Y-layout tri-rotor

Continuous attitude authority; **10 N** per rotor, total **0.82 mg**: a hopper, not a drone.**3.75 kg total** ■ **9.2% leg mass** ■ **20 Nm / leg motor** ■ **500 Hz control**

4. Model - one rigid body, two actuator sets



Net wrench = leg contact + three rotors

$$\begin{bmatrix} \mathbf{F}_{net} \\ \boldsymbol{\tau}_{net} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_c \\ \mathbf{r}_c \times \mathbf{f}_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} \mathbf{d}_i \\ \mathbf{r}_i \times \mathbf{d}_i \end{bmatrix} \mathbf{t}_i$$

Newton-Euler dynamics of the CoM

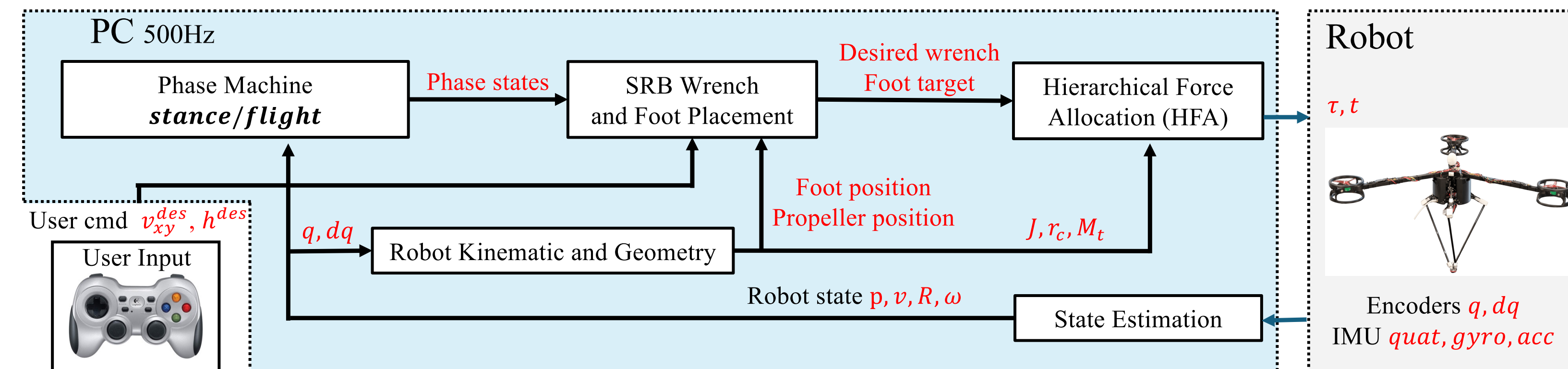
$$m\mathbf{a} = \mathbf{F}_{net} + m\mathbf{g}, \quad \dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{R}^T \boldsymbol{\tau}_{net} - \boldsymbol{\omega} \times \mathbf{I} \boldsymbol{\omega})$$

Paper Fig. 4 - contact force $\mathbf{f}_c$ and thrusts $\mathbf{t}_{1,2,3}$ jointly produce the body wrench.

Symbols (values used in all experiments)

 $\mathbf{f}_c$ ground reaction force (world) $\mathbf{r}_c, \mathbf{r}_i$ arms: CoM → contact, rotor i $\mathbf{t}_i \in [0, 10]$ N thrust along $\mathbf{d}_i = \mathbf{R}\mathbf{e}_3$ $\mathbf{F}_{net}, \boldsymbol{\tau}_{net}$ net force, moment $m=3.75$ kg; $\mathbf{g}$ gravity; $\mathbf{I}$ inertia $\mathbf{R} \in \text{SO}(3)$ attitude; $\boldsymbol{\omega}$ body rate $\mathbf{J}$ leg Jacobian; $\tau_{max}=20$ Nm limit $\mu=0.4$ friction coefficient $p_z, \dot{p}_z$ CoM height, velocity; $h_{des}=0.664$ m l leg length; $l_0=0.464$ m; Δ_{td} threshold k_z, b_z, k_E vertical, k_v, k_r Raibert gains $\mathbf{K}_R, \mathbf{K}_\omega$ attitude PD gains

5. Control - leg-priority hierarchical force allocation



Paper Fig. 5 - implemented 500 Hz architecture: user commands and feedback → planner → two-stage allocator → leg torques + rotor thrusts.

A. Vertical energy hopping (stance)

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + m g p_z, \quad E_{des} = m g h_{des}$$

$$\mathbf{f}_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

B. SO(3) attitude PD (both phases)

$$\mathbf{e}_R = \frac{1}{2} (\mathbf{R}_{des}^T \mathbf{R} - \mathbf{R}^T \mathbf{R}_{des})^\vee$$

$$\boldsymbol{\tau}_b^{des} = -\mathbf{K}_R \mathbf{e}_R + \mathbf{K}_\omega (\boldsymbol{\omega}^{des} - \boldsymbol{\omega})$$

Roll/pitch only; yaw left free. $\boldsymbol{\tau}^{des} = \mathbf{R}^T \boldsymbol{\tau}_b^{des}$.

C. Leg first - constrained least squares

$$\mathbf{f}_{c,xy}^{cmd} = \arg \min_{\mathbf{f}_c} \|(\mathbf{r}_c \times \mathbf{f}_c)_{rp} - (\boldsymbol{\tau}^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|\mathbf{f}_{xy}\|_2 \leq \mu \mathbf{f}_{c,z}^{des}, \quad \|\mathbf{J}^T(-\mathbf{R}^T \mathbf{f}_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

D. Rotors take only the residual

$$\boldsymbol{\tau}_{res}^{cmd} = \boldsymbol{\tau}^{des} - \mathbf{r}_c \times \mathbf{f}_c^{cmd}$$

$$\delta \mathbf{t}^{cmd} = \arg \min_{\delta \mathbf{t}} \|(\mathbf{M}_t \delta \mathbf{t})_{rp} - (\boldsymbol{\tau}_{res}^{cmd})_{rp}\|_2^2$$

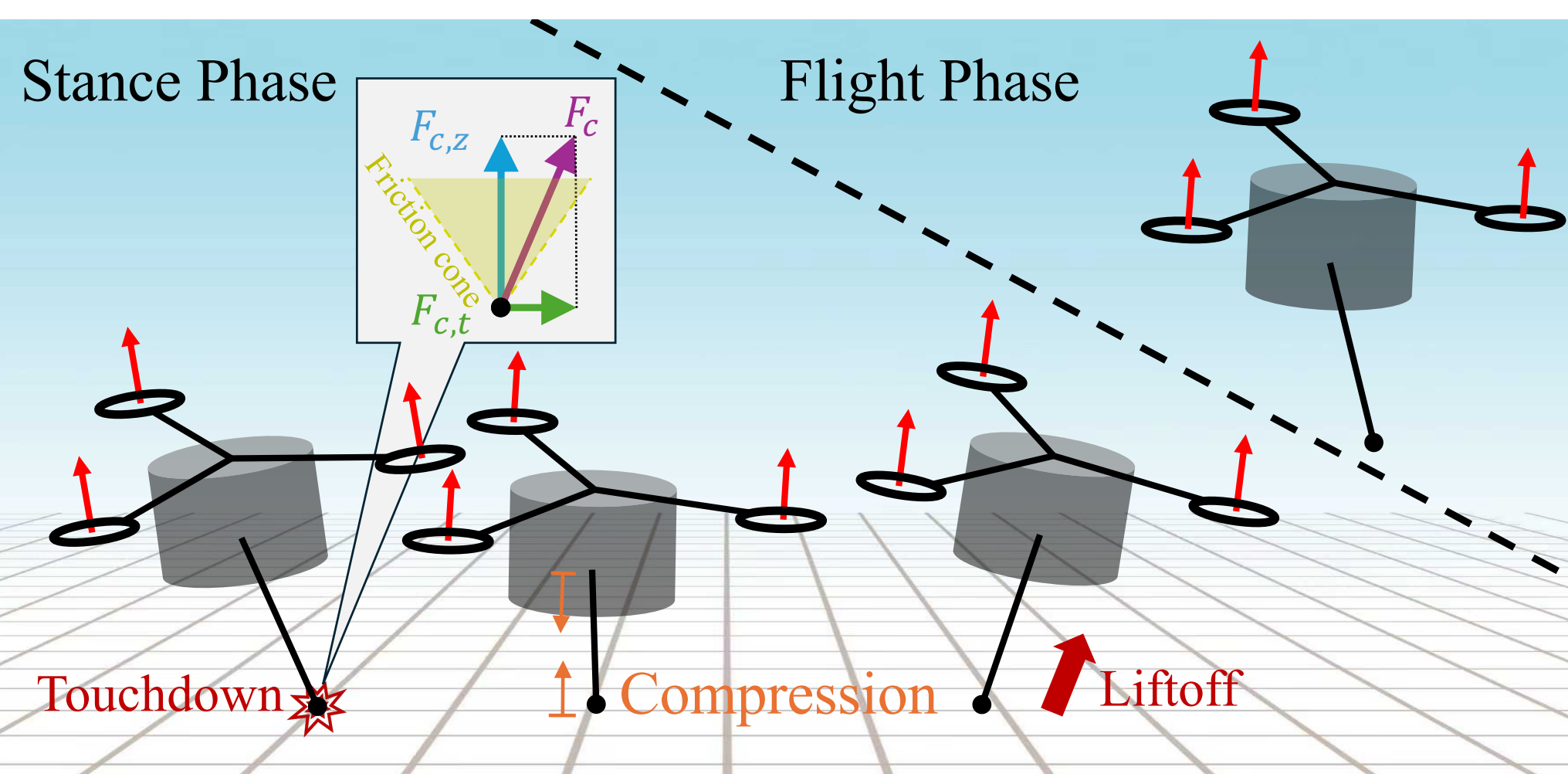
s.t. thrust bounds; $\mathbf{t}^{cmd} = \frac{T_{base}}{3} \mathbf{1}_3 + \delta \mathbf{t}^{cmd}$, $T_{base} = 0.02$ mg.

E. Flight - rotors primary for attitude; Raibert foot placement

 $\mathbf{f}_c = 0$; rotors track $(\boldsymbol{\tau}^{des})_{rp}$; leg tracks $\mathbf{r}_{td,xy}^{des} = k_v \mathbf{v}_{xy} - k_r \mathbf{v}_{xy}^{des}$

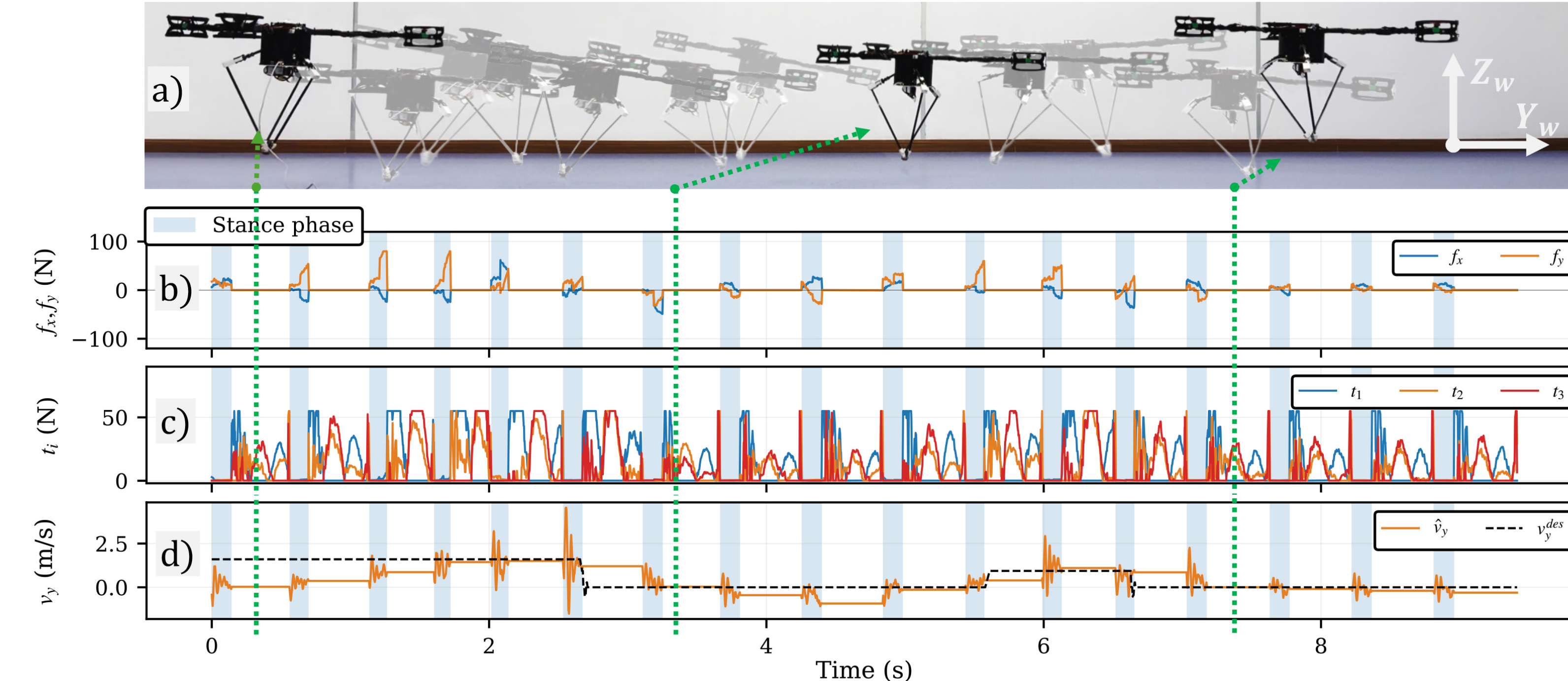
By construction the rotors can only add to - never replace - the leg's contact action.

6. Onehop - event-driven authority handoff

**Touchdown:** $l \leq l_0 - \Delta_{td}$
Leg absorbs impact; energy law drives the next apex.**Liftoff:** $l \geq l_0$
Contact wrench vanishes; rotors primary for attitude.**Handoff**
Events switch authority; no hardware or gain change.

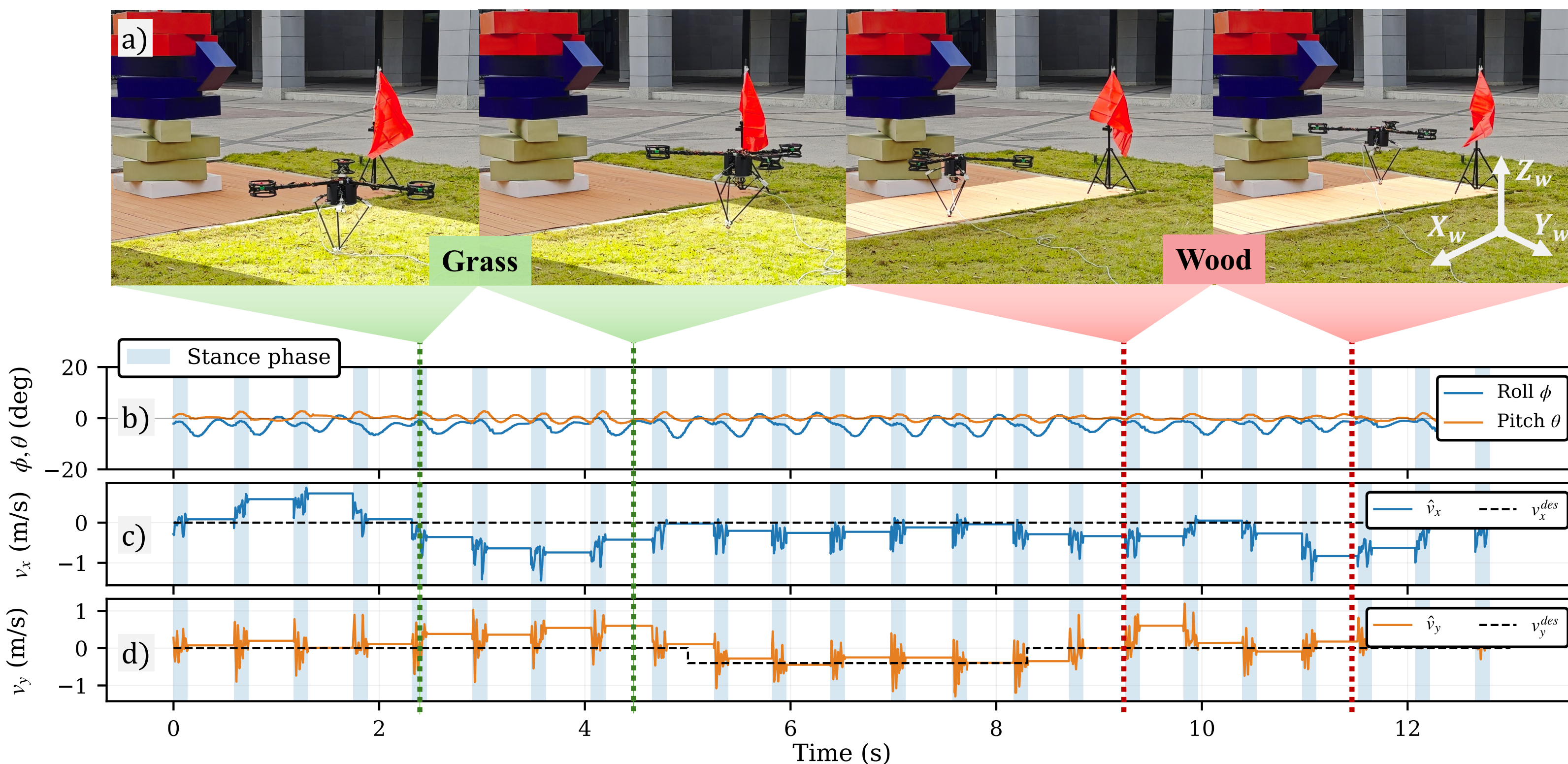
Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

7. Validation - indoor tracking



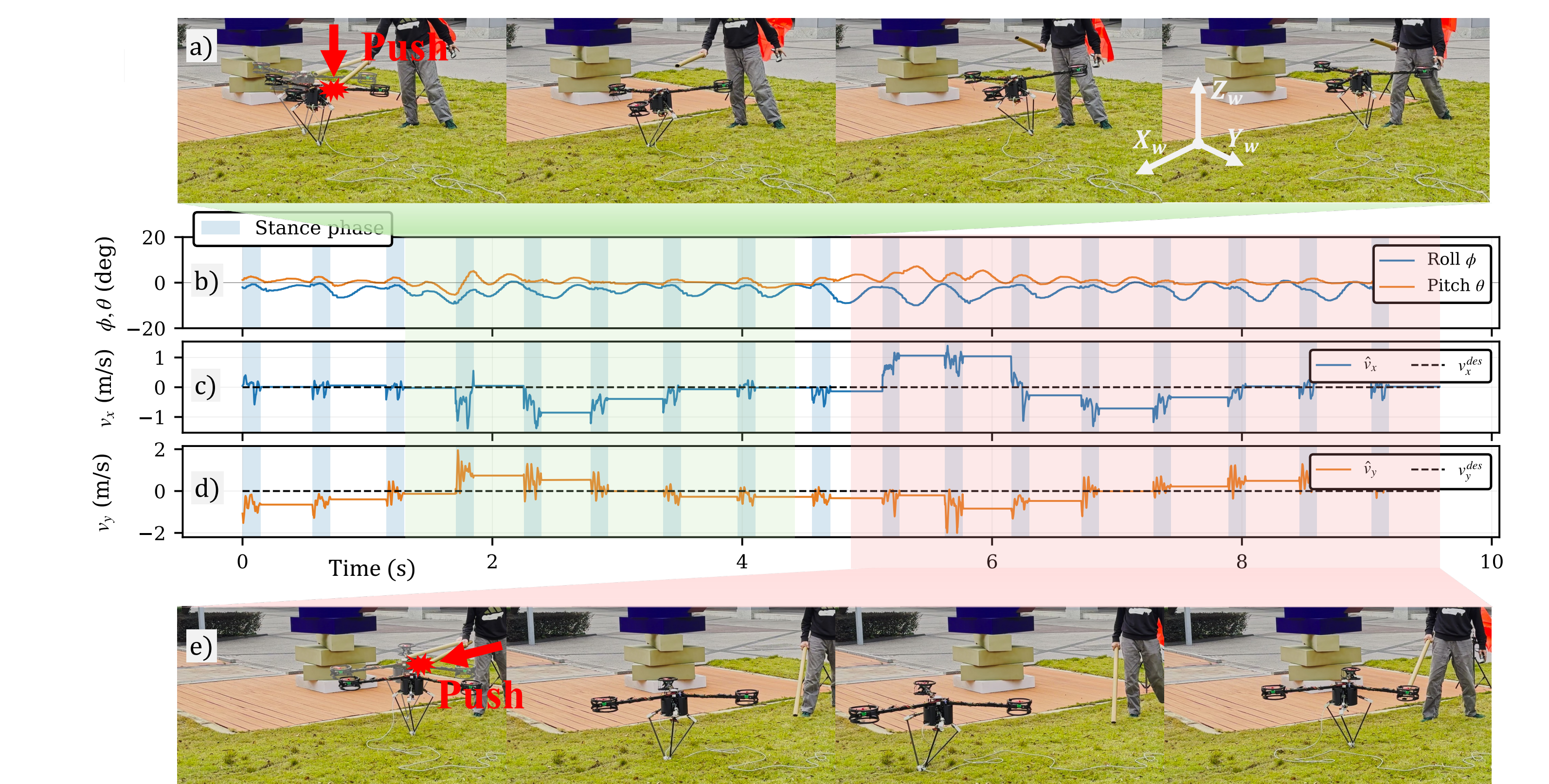
17 hops - 1.6 and 0.94 m/s commands - bounded attitude error through every impact.

8. Validation - grass to wood



23 hops - same controller - no terrain model - no gain retuning.

9. Validation - push recovery



Rotor-arm hit: attitude recovers in 1-2 hops. Main-body hit: lateral velocity recovers in three hops.